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# Rule-Based Chatbot using NLTK and spaCy

## Objective
Build a chatbot that understands a user's question even when it is phrased
differently, by combining **both NLTK and spaCy** in a single NLP pipeline. The
bot answers common questions for an AI/ML training academy.

## How It Works
1. **NLTK cleans the sentence:** it splits the input into words (`word_tokenize`)
   and removes filler words like "the", "is", and "a" (stopwords).
2. **spaCy simplifies the words:** it reduces each word to its root form
   (lemmatization), so "studying", "studies", and "studied" all become "study".
3. **The bot compares and responds:** the cleaned words are matched against the
   keywords of each known topic (intent), and the best match's response is returned.

## Supported Topics (Intents)
| Intent | Example keywords | Response |
|---|---|---|
| Greeting | hello, hi, hey | Greets the user |
| Courses | course, learn, study, class | AI and ML levels offered |
| Fees | fee, price, cost, payment | Directs user to the admin office |
| Goodbye | bye, goodbye, exit, quit | Ends the conversation |

If nothing matches, the bot replies: "Sorry, I didn't understand that. Can you
rephrase?"

## Example
    You: hello
    Bot: Hello! How can I help you today?
    You: how much is the fee
    Bot: Please contact the admin office for current fees and payment plans.

## Limitations and Future Work
- Matching is keyword-based, so it cannot handle context or multi-turn conversation.
- The knowledge base has only 4 intents.
- Next steps: add more intents, use spaCy word vectors or a trained intent
  classifier, and deploy it as a web or WhatsApp bot.

## Tech Stack
Python, NLTK, spaCy (`en_core_web_sm`)
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# How NLTK and spaCy Work Together in My Chatbot

#The task asked me to use **both** NLTK and spaCy, not build two separate chatbots.
#Each one does a different job in the same pipeline, one after the other:

#1. **NLTK cleans the sentence.**
   #It splits the user's question into individual words,
  #then removes filler words like "the", "is", and "a".

#2. **spaCy simplifies the words.**
   #It takes those cleaned words and reduces each one to its root form.
   #For example, "studying", "studies", and "studied" all become "study".

#3. **The bot compares and responds.**
 #  Once the words are cleaned and simplified, the bot compares them
  # against a list of known topics and picks the response that matches best.

#In short:** NLTK tidies up the sentence, spaCy standardizes the words,
#and together they help the bot understand questions even when people
#phrase them differently.
```

```
!pip install nltk spacy
!python -m spacy download en_core_web_sm
```

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Requirement already satisfied: nltk in /usr/local/lib/python3.13/dist-packages (3.9.1)
Requirement already satisfied: spacy in /usr/local/lib/python3.13/dist-packages (3.8.16)
Requirement already satisfied: click in /usr/local/lib/python3.13/dist-packages (from nltk) (8.5.0)
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Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.13/dist-packages (from nltk) (2025.11.3)
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Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.13/dist-packages (from spacy) (3.0.12)
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Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.13/dist-packages (from spacy) (2.0.13)
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Requirement already satisfied: typer<1.0.0,>=0.3.0 in /usr/local/lib/python3.13/dist-packages (from spacy) (0.27.2)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.13/dist-packages (from spacy) (2.1.3)
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Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.13/dist-packages (from spacy) (80.10.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.13/dist-packages (from spacy) (26.3)
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Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests<3.0.0,>=2.13.0->spa
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests<3.0.0,>=2.13.0->spa
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Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.13/dist-packages (from jinja2->spacy) (3.0.3)
Requirement already satisfied: cloudpickle>=3.0 in /usr/local/lib/python3.13/dist-packages (from joblib->nltk) (3.1.2)
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Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.13/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-an
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 46.2 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
import nltk
import spacy
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')

nlp = spacy.load('en_core_web_sm')
stop_words = set(stopwords.words('english'))
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
```

```
knowledge = {
    "when_user_says_hello": {
        "keywords": ["hello", "hi", "hey", "morning", "afternoon"],
        "response": "Hello! How can I help you today?"
    },
    "when_user_asks_about_course": {
        "keywords": ["course", "learn", "study", "subject", "class"],
        "response": "We offer AI and Machine Learning at beginner, intermediate and advanced levels."
    },
    "when_user_asks_about_fee": {
        "keywords": ["fee", "price", "cost", "pay", "payment"],
        "response": "Please contact the admin office for current fees and payment plans."
    },
```

```
    "when_user_says_bye": {
        "keywords": ["bye", "goodbye", "exit", "quit"],
        "response": "Goodbye! Have a great day."
    },
}
```

```
def preprocess(text):
    tokens = word_tokenize(text.lower())
    tokens = [t for t in tokens if t.isalpha() and t not in stop_words]
    doc = nlp(" ".join(tokens))
    return {tok.lemma_ for tok in doc}

for intent in knowledge.values():
    intent["lemmas"] = preprocess(" ".join(intent["keywords"]))

def get_response(user_input):
    user_words = preprocess(user_input)
    best_intent, best_score = None, 0
    for name, intent in knowledge.items():
        score = len(user_words & intent["lemmas"])
        if score > best_score:
            best_intent, best_score = name, score
    if best_intent:
        return knowledge[best_intent]["response"], best_intent
    return "Sorry, I didn't understand that. Can you rephrase?", None
```

```
print("Bot: Hi! Ask me a question (type 'bye' to exit).")
while True:
    user = input("You: ")
    reply, intent = get_response(user)
    print("Bot:", reply)
    if intent == "when_user_says_bye":
        break
```

```
Bot: Hi! Ask me a question (type 'bye' to exit).
You: hello
Bot: Hello! How can I help you today?
You: how can i pay
Bot: Please contact the admin office for current fees and payment plans.
You: but what can actually learn
Bot: We offer AI and Machine Learning at beginner, intermediate and advanced levels.
You: okay goodbye
Bot: Goodbye! Have a great day.
```